

Project Draft Assignment

This is the assignment for your *draft project*, to be turned in as a report on your results to date for your project. Read the sections below, as a guide for the tasks needed to complete your project, then report on your current results for each task:

- What you have to show -- intermediate results
- What you need to do next
- What if anything is blocking your progress.

You do not have to do these tasks in order. You can be concise, and skip over sections that are yet to be done.

The tasks are built around the 6 areas of Decision Quality -- the criteria by which your final project will be evaluated. *Although your project will have to address all six criteria, you should select one area of the six that your project exemplifies.* Under the following list, pick one that is primary.

Task: Appropriate Framing: Solve the Right Problem

What is your primary decision? - in terms that meet the requirement for what is a decision. Based on your decision, what is the *scope* of the problem -- aspects that should be included and should be outside of the scope.

Task Clear Values and Tradeoffs: What does the decision maker want to achieve?

What values need to be considered to define the consequences of the decision? There are inevitably tradeoffs to be made for any choice, what are they? Are there ethical concerns? How do you consider monetary value, risk and time preference? What are the subjective values you will include, and how, and from whom are you eliciting them? You should draw up the network of value nodes for your CDN.

Task Creative Alternatives: What range of options address the decision-makers goals?

For your primary decision, enumerate the alternatives facing the decision-maker. Can you expand the set alternatives that might be more likely to meet the decision-maker's goals? If it makes sense in your case, are there a sequence of subsequent choices to make adjustments after the primary decision is made?

Task Relevant and Reliable Information: What uncertainties affect the decision outcomes?

Define the variables for which probability distributions need to be defined. The distinctions by which they are defined should pass the "clarity test." What variables are relevant to the model's values? What variables are observable, and could be used to compute VOI on relevant variables of interest? Include any test decisions for observables. Draw up the Bayes network sub-model for your CDN. Indicate how you will elicit probability distributions for them. Will you refer to any experts outside of your project team for information?

Task Sound Reasoning: How is your model solved to generate recommendations?

Assemble the decision, value, and probability sub-models into a consistent CDN. What computational steps are needed to analyze it? Will you need to use software for this? What can you discover by running sensitivity analysis -- e.g. by varying aspects of the model, such as information observed at the time of the decision -- to see how it might change model recommendations?

Task Commitment to Action: Are the model's recommendations credible?

Does your analysis make a clear case to the decision-maker to inform their decision? How do you explain convincingly to the decision maker -- the basis for your recommendation using the decision model that you've built? How do you visualize the results?